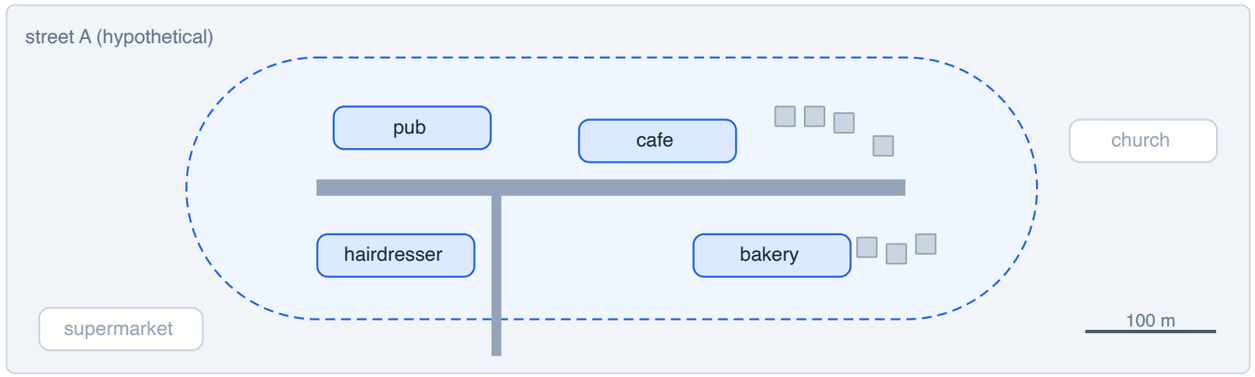
Stage 1 on a map: which street is the café from?

A hypothetical example of the stage-1 quizzes, drawn as streets on a map. The numbers are in the main stage-1 diagram (*cityfm_stage1_train_test*).

within 100 m of the street · cafe counted for the street · church too far away, not counted · cafe the picked object · made more similar · made less similar

A What CityFM knows about a street

CityFM never sees the map itself. Each street is described only by a list of the things within about 100 m of it, written as their types. Names are left out, and houses are counted and replaced by a single item.



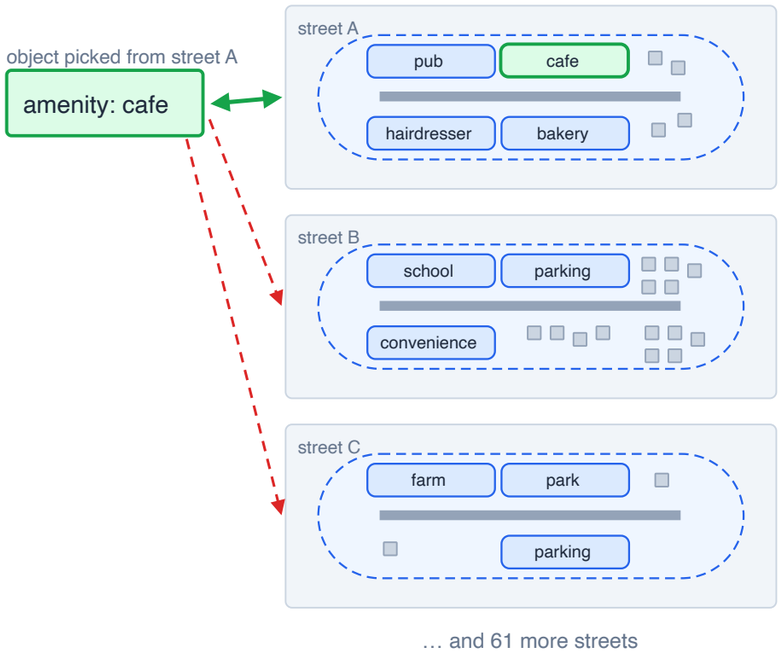
what CityFM reads for street A

- amenity: pub
- amenity: cafe
- shop: hairdresser
- shop: bakery
- building: few houses

The church and the supermarket are not in the list because they are more than 100 m away. The seven houses become one item, *building: few houses*.

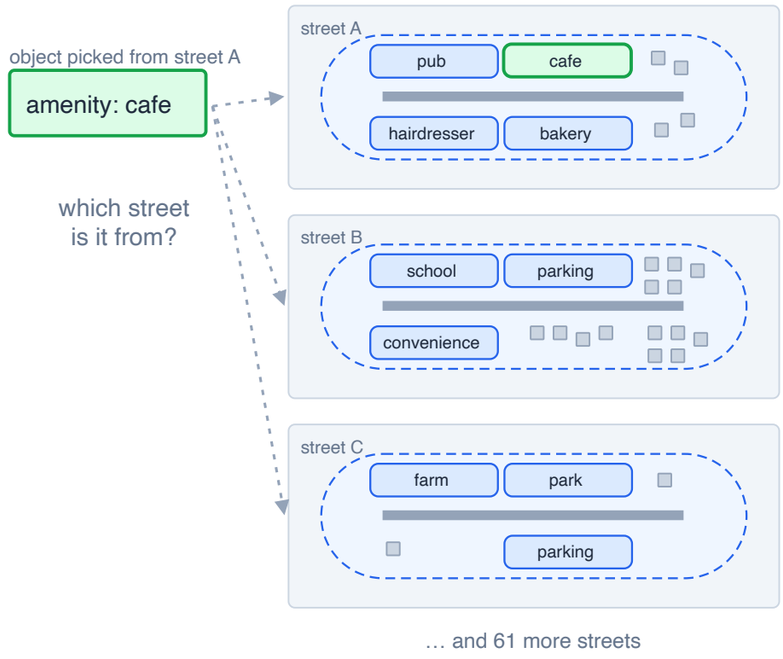
1 Training

One object is picked from street A, here the café. The model is adjusted so that the café is more similar to street A than to the other 63 streets. The café is still on street A, so spotting it is enough to win.



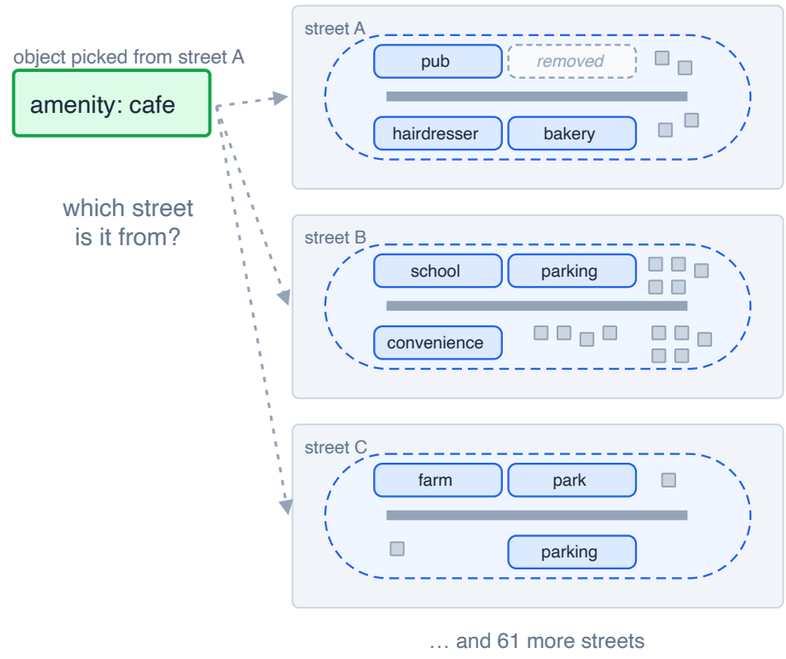
2 Test 1 · Café left in

The finished model is asked which of the 64 streets the café comes from. As in training, the café is still in street A's list, so a model can answer by looking for *amenity: cafe*.



3 Test 2 · Café removed

The café is taken off street A before asking. The model must judge from what is left: is a café more likely next to a pub, a bakery and a hairdresser, or next to a school, or a farm?



Why there are two tests. In training and test 1 the café is still on its street, so a model can score well by matching the word *cafe* without learning anything about places. Test 2 removes that shortcut. A model can only pass it if it has learned which kinds of place are usually found near each other, which is what stage 1 is meant to teach.

The streets and objects are invented for illustration. In the real data the 100 m is measured from the points that make up each mapped road segment, and each quiz uses 64 streets drawn at random from the city.